Shivam Pawar

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Education

Manipal University Jaipur — **August 2024 – Present**
B.Tech in Computer Science and Engineering(AI & ML) — *CGPA-9.07*

Projects

Asthma Prediction System with Explainable AI | *Python, ML, SHAP, LIME* — **September 2025-March 2026**

- Developed a machine learning model to predict asthma risk using clinical and environmental data.
- Integrated explainability techniques (SHAP and LIME) to generate interpretable predictions for better understanding.
- Designed a structured pipeline for data processing, model training, and evaluation.
- Generated detailed explanations using a language model to improve clarity of results.

Customer Segmentation using Advanced Unsupervised Learning | *Scikit-learn, Pandas, Matplotlib* — **January 2026**

- Engineered an end-to-end customer segmentation pipeline leveraging multiple clustering algorithms including K-Means, Hierarchical Clustering, and DBSCAN.
- Transformed raw transactional data through extensive preprocessing, feature engineering, and outlier handling to improve data quality and model reliability.
- Optimized clustering performance using the Elbow Method and Silhouette Score, enabling data-driven selection of optimal cluster configurations.
- Uncovered meaningful customer patterns and behavioral segments through exploratory data analysis and advanced visualizations.

Healthcare Prediction using Machine Learning Models | *Python, Scikit-learn, Pandas* — **December 2025**

- Developed and evaluated classification models including Logistic Regression and Boosting algorithms on healthcare datasets.
- Performed data preprocessing, feature selection, and handled missing values to improve model accuracy and reliability.
- Analyzed model performance using evaluation metrics and comparative analysis to identify optimal predictive models.

Publications

LLMs as Explainable Layer for ML-based Asthma Prediction Model | *Manuscript in Preparation* — **April 2026**

- Research paper focused on interpretable machine learning for healthcare prediction.
- Manuscript currently in preparation, focusing on validation and refinement of experimental results.

Technical Skills

Languages: Python, C, SQL, MATLAB
Machine Learning: Supervised Learning, Unsupervised Learning, Ensemble Methods (Boosting), Data Preprocessing
Libraries/Frameworks: Scikit-learn, Pandas, NumPy, Seaborn, Matplotlib
Tools: Git, GitHub, VS Code
Operating Systems: Windows, Linux

Relevant Coursework

- Data Structures & Algorithms
- Operating Systems(OS)
- Object-Oriented Programming
- Design & Analysis of Algorithms
- Database Management Systems
- Principles of Artificial Intelligence
- Statistics & Probability
- Regression Analysis & Forecasting
- Calculus & Matrices
- Cloud Computing
- Software Engineering & Project Management

Certifications

Machine Learning with Python — **December 2025**
IBM, Coursera (View Certificate)

Generative AI for Software Development — **December 2025**
DeepLearning.AI, Coursera (View Certificate)

Python Essentials 1 & 2 — **October 2025**
Cisco Networking Academy (View Certificate)